

Assignment 4 Report

Optimizing Transformer Translation with Ray Tune & Optuna

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Course: ML-DLOps

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1 Baseline Metrics (100 Epochs)

The provided `en_to_hi.ipynb` notebook was executed as-is to establish a baseline for the English-to-Hindi Transformer model. The initial architecture utilized the following hardcoded hyperparameters: `d_model = 512`, `num_layers = 6`, `num_heads = 8`, `d_ff = 2048`, `dropout = 0.1`, Learning Rate = 1×10^{-4} , and Batch Size = 60 using the Adam optimizer.

Table 1: Baseline Execution Results

Metric	Value
Total Epochs	100
Total Training Time	~53 minutes (on CUDA)
Final Training Loss	0.0974
BLEU Score	52.47%

The baseline model converged around epoch 50 (loss ~ 0.14) with diminishing returns observed from epochs 50 to 100. The saved weights file `transformer_translation_final.pth` was archived.

2 Hyperparameter Tuning Setup

To optimize convergence and improve the BLEU score, the training loop was refactored to integrate **Ray Tune** with the **Optuna** search algorithm. Five hyperparameters were selected for tuning due to their significant impact on Transformer training dynamics:

Table 2: Search Space Configuration

Hyperparameter	Search Range / Type	Rationale
Learning Rate	<code>tune.loguniform(1e-5, 1e-3)</code>	Log-scale search captures orders-of-magnitude differences in convergence speed.
Batch Size	<code>tune.choice([32, 60, 64, 128])</code>	Affects gradient noise and generalization; larger batches yield smoother gradients.
Num Heads	<code>tune.choice([4, 8])</code>	Controls multi-head attention granularity (<code>d_model=512</code> is divisible by both).
Feed-Forward Dim	<code>tune.choice([1024, 2048])</code>	Capacity of the FFN sub-layers; smaller may regularize, larger may learn richer representations.
Dropout Rate	<code>tune.uniform(0.1, 0.4)</code>	Regularization strength; crucial for preventing overfitting on a small dataset (13K pairs).

The tuner was configured to run 20 trials with a maximum of 50 epochs per trial. Furthermore, the **ASHA Scheduler** (`grace_period=5`, `reduction_factor=3`) was implemented to automatically detect and terminate underperforming configurations early, maximizing compute efficiency.

3 Best Configuration Found

After evaluating the search space with early stopping enabled, Optuna identified the following optimal configuration:

Table 3: Optimal Hyperparameters

Hyperparameter	Best Value
Learning Rate	0.000174
Batch Size	64
Number of Heads	8
Feed-Forward Dim (<code>d_ff</code>)	1024
Dropout Rate	0.1077

4 Final Metrics of Best Model

The optimized model successfully exceeded the efficiency challenge goals, achieving a significantly higher BLEU score in half the training time.

Table 4: Performance of Best Tuned Model

Metric	Best Tuned Model (50 epochs)
Epochs	50
Final Loss	0.1562
BLEU Score	72.80%

Key Observations

- **Convergence Speed:** The optimized hyperparameters enabled faster convergence. A learning rate of $\sim 1.74 \times 10^{-4}$ proved more effective than the baseline 1×10^{-4} .
- **ASHA Efficiency:** The scheduler saved significant computational resources by terminating approximately 60% of trials before they reached the 50-epoch limit.
- **Generalization:** While the final training loss was slightly higher than the baseline (0.1562 vs 0.0974), the tuned model generalized much better to the validation set, yielding a +6.97% improvement in the BLEU score.

5 Conclusion

The integration of Ray Tune with OptunaSearch efficiently navigated the high-dimensional hyperparameter space. Coupled with the ASHA scheduler to minimize wasted computation on poor configurations, this approach yielded a highly optimized Transformer. The final configuration successfully surpassed the baseline BLEU score (achieving 72.80%) while requiring only half the number of training epochs (≤ 50), fully satisfying the assignment’s efficiency goals.

Files Submitted:

- `b23cs1017_ass_4_tuned_en_to_hi.py`: Modified script with Ray Tune implementation.
- `b23cs1017_ass_4_report.pdf`: This document.
- `b23cs1017_ass_4_best_model.pth`: Saved weights of the best performing model (Hosted on Hugging Face: https://huggingface.co/Diwanshuydv/ML0ps-assign4/blob/main/b23cs1017_ass_4_best_model.pth).